

Milestone 3

Proof of Achievement Report

Project Catalyst #1200045: MPC as a Layer 2 Service to Cardano

Profila AG x Partisia Blockchain

Date: 2026-03-06

Network: Partisia Blockchain Testnet

1. Executive Summary

Milestone 3 delivers the secret input submission flow for privacy-preserving MPC computation. A synthetic test dataset of 50 users (representing Profila platform data) is defined, and 5 secret values are submitted as real on-chain ZK transactions to the Partisia Blockchain testnet contract. No individual user data is ever exposed - only the aggregate MPC result is visible.

All 5 acceptance criteria are met.

2. Acceptance Criteria - Evidence Matrix

#	Criterion	Status	Evidence
1	Dataset defined as data input	PASS	Section 3 - Profila test data (50 users)
2	PDF documentation in English	PASS	This document
3	Data type/definition + secret sharing	PASS	Sections 3-4 - types + submission flow
4	YouTube video of process	PASS	youtu.be/psm2IMU7n1o
5	Demo on PBC testnet	PASS	Section 5 - 5 on-chain ZK transactions

3. Dataset Definition - Profila Test Data

3.1 Data Source

The dataset consists of 50 synthetic test users representing typical Profila platform data. This is personal data and demographics which require analysis in a privacy-preserving manner. The data source is defined as test data from the Profila platform.

Dataset file: test-data/profila_test_users.json

Dataset ID: profila_test_v1

3.2 Data Type Definition

Each user record contains the following fields:

Field	Type	Secret?	Description
user_id_hash	string	No	SHA-256 hash of synthetic user ID
age	number	YES	User age (14-75) - MPC secret input
survey_q1_answer	boolean	YES	Survey response - MPC secret input

country_code	string	No	ISO country code (CH, DE, GB, US, FR, AU)
profilia_consent	boolean	No	Consent flag (always true for PoC)

3.3 MPC Query Types

Query Type	Secret Field	PBC Type	Computation
age_threshold	age	Sbi64	Count where value > 18
survey_match	survey_q1_answer	Sbi64	Sum all values (0 or 1)

3.4 Dataset Distribution

50 users total:

- 35 users over age 18 (70%), 15 users age 18 or under (30%)
- 28 users with survey_q1_answer = true (56%), 22 false (44%)
- 6 countries: CH, DE, GB, US, FR, AU
- Note: Synthetic data only - no real PII

4. Secret Sharing Process

4.1 How Secrets Are Shared into the PBC Contract

The secret submission process uses the official Partisia Blockchain TypeScript SDK. Each user's private value (age or survey answer) is encrypted and secret-shared by PBC's ZK MPC infrastructure across 4 MPC nodes. No single node ever sees an individual value.

Process steps:

1. Load test user record from Profila dataset
2. Extract the secret field (age for age_threshold, survey answer for survey_match)
3. Serialize the secret as Sbi64 (signed 64-bit integer) via BitOutput
4. Encrypt the secret for each MPC engine's public key via RealZkClient
5. Package encrypted shares into a single PBC blockchain transaction
6. Submit the transaction on-chain (submit_secret_value, shortname 0x40)
7. MPC nodes verify and accept the shares asynchronously

4.2 Privacy Guarantees

The submission log contains ONLY: user_id_hash, pbc_tx_id, query_type, timestamp.

Zero raw ages or survey answers are ever logged, stored, or transmitted in plaintext.

The contract state shows only aggregate fields - no individual values visible.

The PBC browser Secret Data tab shows computation status only - no individual ZK variables.

5. On-Chain Evidence - PBC Testnet

5.1 Secret Input Transactions

5 real on-chain ZK secret-input transactions submitted to PBC testnet:

#	PBC Transaction Hash	PBC Sender
1	7507fd18ef779c459c7248af32bee6e660f6b4ad...	0043d3baeb08c94e8402...
2	519859f57ccc1acd44ed73fe80ae2a3c051b15ec...	0077475a2deceb9c073e...
3	407423fd9996067e8c4b5de9390bdad6c14c1da2...	0084400b5b22b47c2414...
4	4bfe3b1466c68f8b2e73376b35677ed2592ffd85...	00d220baae1d73526743...
5	e4c158dda054b2fc9ba17a38a7fb4f8373c4720e...	00d85675808c74ccfebb...

5.2 Verification

PBC Contract on testnet:

<https://browser.testnet.partisiablockchain.com/contracts/039ed9214602f2a93eb05411f852fa78a476607634>

Full submission log with all TX hashes: relay/secret-submissions-log.json

Privacy verification evidence: relay/m3-state-evidence.json

5.3 MPC Computation Result

After secret submission, the MPC computation was triggered and completed:

- Query: age_threshold (count users with age > 18)
 - Result: 5 (all 5 submitted test users had age > 18)
 - Computation method: MPC across 4 PBC ZK nodes
 - No individual data exposed - aggregate result only
-

6. Repository

GitHub: https://github.com/Profila/Aiken-Partisia_MPC

Test dataset: test-data/profila_test_users.json

Secret submission script: relay/submit_secrets.ts

Submission log: relay/secret-submissions-log.json

Privacy evidence: relay/m3-state-evidence.json

M3 report: docs/evidence/milestone-3-report.json

7. Video Demo

M3 Demo: <https://youtu.be/psm2IMU7n1o>

Combined M1-M4 Demo: <https://youtu.be/jwwfdTSUS2w>

Profila AG x Partisia Blockchain
Project Catalyst Fund 13 - Proposal #1200045